

MATH 350-2 Advanced Calculus

W.R. Casper

Department of Mathematics
California State University Fullerton

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Outline

- 1 Real Analysis Lecture 11
 - Limits of sequences

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Warm-up

Definition

The **Oxford dictionary metric space** is the set M of all words in the Oxford English dictionary, where the metric is the discrete metric.

Example:

$$d(\text{fish}, \text{bird}) = 1, \quad d(\text{apple}, \text{apple}) = 0$$

Prove that for any word $w \in M$

$$d(\text{fire}, \text{ice}) \leq d(\text{fire}, w) + d(w, \text{ice})$$

Sequences

Definition

A **sequence** of real numbers is a function $x : \mathbb{N} \rightarrow \mathbb{R}$.

A **sequence** of points in $\mathbb{R}^n$ is a function $x : \mathbb{N} \rightarrow \mathbb{R}^n$

A **sequence** of points in a metric space (M, d) is a function $x : \mathbb{N} \rightarrow M$

Notation:

$$x_n = \text{formula}, \quad x(n) = \text{formula}, \quad x = (a, b, c, \dots)$$

Examples:

- $x_n = 1/n$ is a sequence in $\mathbb{R}$
- $x_n = (1/n, 1/n^2)$ is a sequence in $\mathbb{R}^2$
- $x = (\text{cat}, \text{dog}, \text{cat}, \text{dog}, \dots)$ is a sequence in the Oxford dictionary metric space

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Metric space version:

$$\forall \epsilon > 0, \quad \text{there exists } N \in \mathbb{Z}_+ \text{ such that } d(x_n, L) < \epsilon \quad \forall n \geq N.$$

We say $\{x_n\}$ **converges** to L and L is the **limit of the sequence**.

Example

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Then $n > N$ implies

$$d(x_n, 0) = |1/n - 0| = 1/n < 1/N < \epsilon.$$

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Then $n > N$ implies

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Suppose that it converges to a number L .

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Then for all $\epsilon > 0$, there must exist an $N \in \mathbb{Z}_+$ such that $d(x_n, L) < \epsilon$ for all $n \geq N$.

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Take $\epsilon = 1$ and choose $N \in \mathbb{Z}_+$ such that $d(x_n, L) < 1$ for all $n \geq N$.

With the discrete metric $d(x_n, L) < 1$ implies $x_n = L$.

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With the discrete metric $d(x_n, L) < 1$ implies $x_n = L$.

This means $x_n = 1/n = L$ for all $n \geq N$, which is a contradiction!



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This is a contradiction.



Existence of limits

Question

When do limits *exist*?

- necessary condition: bounded range
- sufficient condition: monotone sequences in $\mathbb{R}$
- sufficient condition: Cauchy in a complete space

Monotone sequences

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In any of these cases, we call the sequence **monotone**.

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A monotone increasing sequence $\{x_n\}$ of real numbers which is bounded above converges to $L = \sup\{x_1, x_2, \dots\}$.

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A monotone decreasing sequence $\{x_n\}$ which is bounded below converges to $L = \inf\{x_1, x_2, \dots\}$.

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Let $\epsilon > 0$.

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Then $L - \epsilon$ is not a supremum of X , so there exists $x_N \in X$ such that $x_N > L - \epsilon$.

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Then $L - \epsilon$ is not a supremum of X , so there exists $x_N \in X$ such that $x_N > L - \epsilon$.

Since the sequence is increasing, for any $n > N$, we have that $L - \epsilon < x_N \leq x_n < L$.

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Then $L - \epsilon$ is not a supremum of X , so there exists $x_N \in X$ such that $x_N > L - \epsilon$.

Since the sequence is increasing, for any $n > N$, we have that $L - \epsilon < x_N \leq x_n < L$.

It follows that

$$d(x_n, L) = |x_n - L| = L - x_n < \epsilon.$$

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$$d(x_n, L) = |x_n - L| = L - x_n < \epsilon.$$

Since $\epsilon > 0$ was arbitrary, this proves x_n converges to L . □

Existence of limits

Theorem

If a sequence $\{x_n\}$ in a metric space (M, d) converges to a value $L \in M$, then the range

$$X = \{x_1, x_2, \dots\} \subseteq M$$

is a bounded set and L is an adherent point of X .

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When the range of a sequence is bounded, we call the sequence $\{x_n\}$ bounded.

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Then there exists $N \in \mathbb{Z}_+$ with $d(x_n, L) \leq 2026$ for all $n \geq N$.

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If we take

$$R = \max\{d(x_1, L), d(x_2, L), \dots, d(x_{N-1}, L), 2026\},$$

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then $d(x_n, L) \leq R$ for all N .

Therefore $X \subseteq B_M(L, R)$ and in particular X is bounded. □

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Theorem

If S is a subset of a metric space (M, d) and $L \in M$ is an adherent point of S , then there exists a sequence $\{x_n\}$ of elements of S which converges to L .

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Then for all $n \in \mathbb{Z}_+$, the ball $B_M(x, \frac{1}{n})$ contains at least one element of S .

For each $n \in \mathbb{Z}_+$, choose $x_n \in S \cap B_M(x, \frac{1}{n})$.

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We claim $\lim x_n = L$.

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To see this, let $\epsilon > 0$.

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Then choose $N \in \mathbb{Z}_+$ with $N > 1/\epsilon$.

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To see this, let $\epsilon > 0$.

Then choose $N \in \mathbb{Z}_+$ with $N > 1/\epsilon$.

It follows that for all $n \geq N$

$$d(x_n, L) < 1/n < 1/N < \epsilon.$$

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Suppose L is an adherent point of S .

Then for all $n \in \mathbb{Z}_+$, the ball $B_M(x, \frac{1}{n})$ contains at least one element of S .

For each $n \in \mathbb{Z}_+$, choose $x_n \in S \cap B_M(x, \frac{1}{n})$.

We claim $\lim x_n = L$.

To see this, let $\epsilon > 0$.

Then choose $N \in \mathbb{Z}_+$ with $N > 1/\epsilon$.

It follows that for all $n \geq N$

$$d(x_n, L) < 1/n < 1/N < \epsilon.$$

Since $\epsilon > 0$ was arbitrary, this proves convergence. □